



PARSHVANATH CHARITABLE TRUST'S
A. P. SHAH INSTITUTE OF TECHNOLOGY
Department of Information Technology
(NBA Accredited)



Semester: VII
Academic Year: 2026-27
Class / Branch: BE IT-C
Subject: Recent Open Source Lab
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Name of Student: Shreya Sonawale
Student ID: 23104134

EXPERIMENT NO. 03

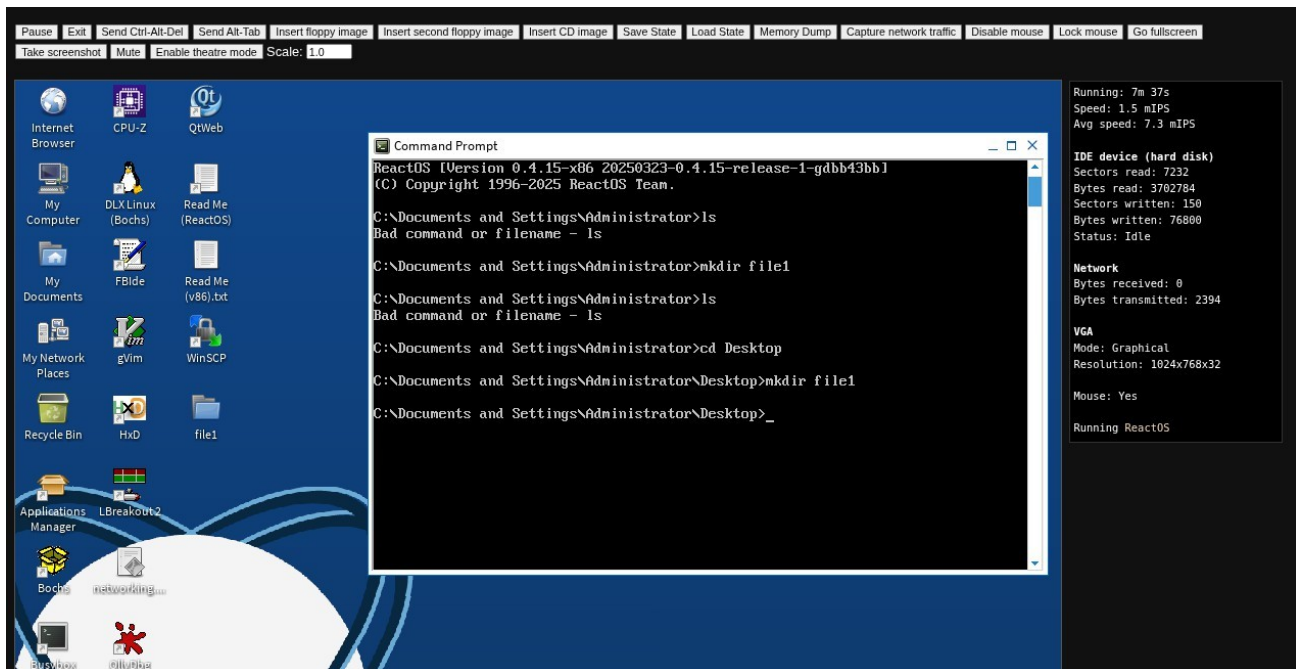
Aim: To explore various open-source operating systems like ReactOS, ChromeOS, includeOS and understand steps for contributing in ReactOS.

Steps to Contribute in ReactOS :





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```
Command Prompt

ReactOS IP Configuration

Ethernet Adapter: Network Connection 3

    Connection-specific DNS Suffix. . . : 
    IP Address. . . . . : 0.0.0.0
    Subnet Mask . . . . . : 0.0.0.0
    Default Gateway . . . . . : 10.5.0.1

C:\Documents and Settings\Administrator\Desktop>ipconfig

ReactOS IP Configuration

Ethernet Adapter: Network Connection 3

    Connection-specific DNS Suffix. . . : 
    IP Address. . . . . : 169.254.2.44
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 10.5.0.1

C:\Documents and Settings\Administrator\Desktop>ping 10.5.73.229

Pinging 10.5.73.229 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.5.73.229:
    Packets: Sent = 4, Received = 0, Lost 4 (100% loss),
```

```
Command Prompt

Packets: Sent = 4, Received = 0, Lost 4 (100% loss),

C:\Documents and Settings\Administrator\Desktop>systeminfo
Host Name:                REACTOS-PY4FDIP
OS Name:                  ReactOS
OS Version:               5.2.3790 Service Pack 2 Build 3790
OS Build Type:             Uniprocessor Checked
Registered Owner:         Owner
Registered Organization:
Warning! Cannot query ProductId. Last error: 0, type: 0.
Product ID:
Install Date:             6/11/2025, 1:24:56 PM
System Up Time:           0 Days, 0 Hours, 22 Minutes, 9 Seconds
System Manufacturer:
System Model:
System type:              X86-based PC
Processor(s):             1 Processor(s) Installed.
                          [01]: x86 Family 15 Model 6 Stepping 3 GenuineIn
tel
Warning! Cannot query SystemBiosVersion. Last error: 2, type: 0.
BIOS Version:
BIOS Date:                04/01/14
ReactOS Directory:        C:\ReactOS
System Directory:         C:\ReactOS\System32
Boot Device:              \Device\HarddiskVolume1
System Locale:             en-us:English (United States)
Input Locale:             en-us:English (United States)
Time zone:                (GMT-08:00) Pacific Time (US & Canada)
Total Physical Memory:    511 MB
Available Physical Memory: 367 MB
Virtual Memory: Max Size: 2 047 MB
Virtual Memory: Available: 2 027 MB
Virtual Memory: In Use:   20 MB
Page File Location(s):    C:\pagefile.sys
Network Card(s):          1 Installed,
```



```
C:\Documents and Settings\Administrator\Desktop>hostname  
REACTOS-PY4FDIP  
  
C:\Documents and Settings\Administrator\Desktop>_
```



sonawaleshreya288-prog 1m ago

Last edited by sonawaleshreya288-prog ▾ ⋮

Step 1: Continuous Integration with GitHub Actions Objective

Set up a local Python development environment, run automated unit tests, and generate a code coverage report. This demonstrates the basics of Continuous Integration (CI) by verifying that the application works correctly before making changes.

Steps

Open the repository in GitHub Codespaces.

Review the project files:

src/calculations.py

tests/calculation_tests.py

Open the terminal and create a virtual environment:

```
python -m venv .venv/calculations
```

```
source .venv/calculations/bin/activate
```

Install the required dependencies:

```
pip install -r requirements.txt
```

```
pip install pytest coverage pytest-cov
```

Run the tests with coverage:

```
pytest --cov=src --verbose
```

Review the test results and coverage summary.

Conclusion

Thus We have studied different Open source Operating System and steps to contribute in ReactOS.



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